

**사용 도구: Cisco
Packet Tracer**

네트워크 구축 및 DHCP · SSH 구성

정현성

010.2528.7127

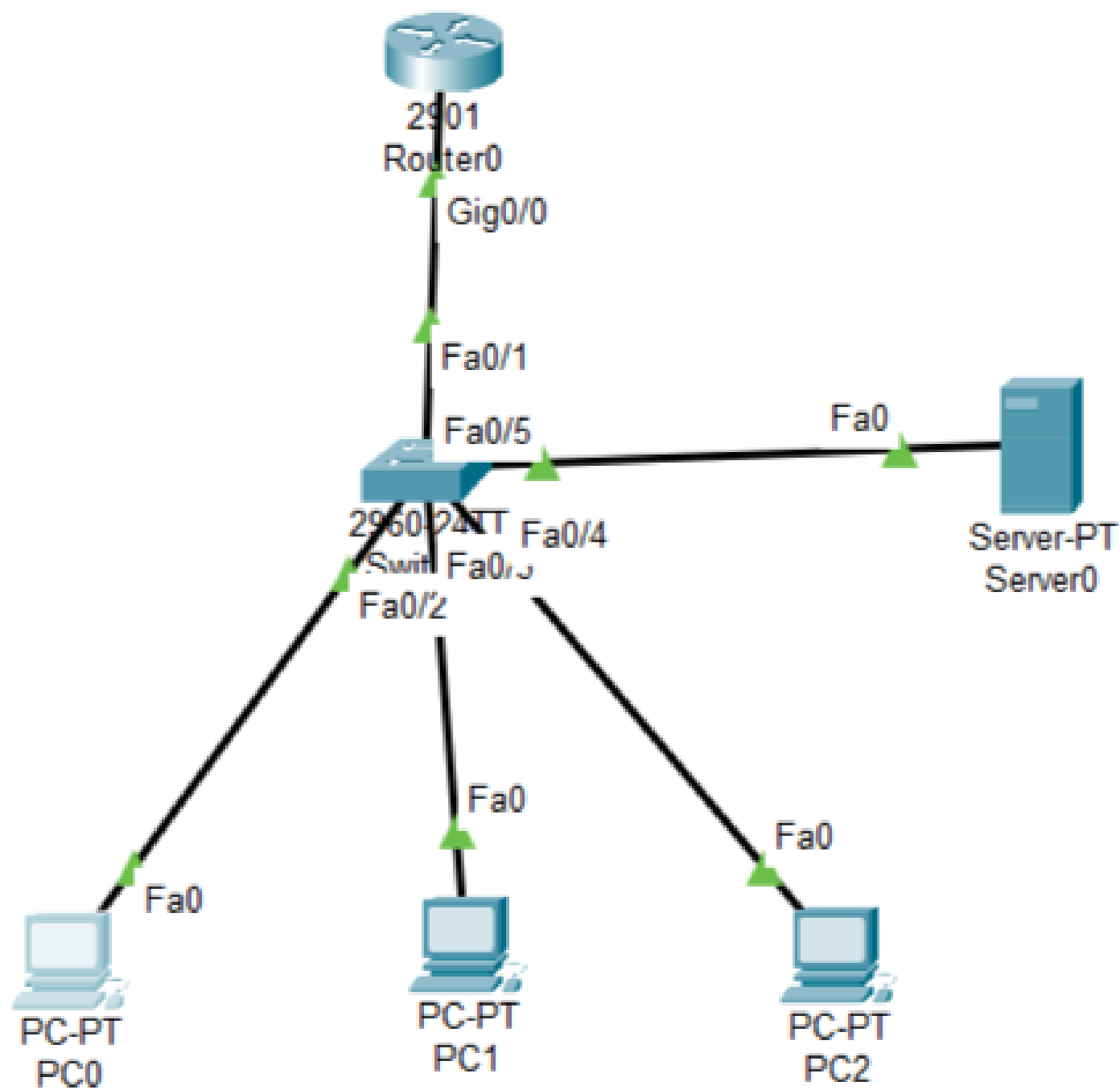
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프로젝트 개요

- 01 Router, Switch, PC, Server를 활용한 기본 LAN 환경 구축
- 02 DHCP 기반 자동 IP 할당 구성
- 03 Server 고정 IP 구성

- 04 SSH(Port 22) 원격 접속 환경 구축
- 05 PC ↔ Server ↔ Router 간 통신 검증

네트워크 구성도



네트워크 환경

- 장비 : 역할
- Router : Gateway / DHCP Server / SSH Server
- Switch : Layer2 스위칭
- PC0~2 : DHCP Client
- Server : Static IP Server

IP 주소체계

장비	IP	주소방식
Router	G0/0 192.168.10.1	Static
Server	192.168.10.100	Static
PC0~2	192.168.10.x	DHCP

서브넷 마스크 : 255.255.255.0

```
!  
interface GigabitEthernet0/0  
  ip address 192.168.10.1 255.255.255.0  
  duplex auto  
  speed auto  
!
```

Router 인터페이스 설정

기능

- Router 인터페이스 활성화
- Gateway ip 설정

명령어

enable

configure terminal

interface g0/0

ip address 192.168.10.1 255.255.255.0

no shutdown

end

copy run start

```
:  
!  
ip dhcp excluded-address 192.168.10.1  
ip dhcp excluded-address 192.168.10.100  
!  
ip dhcp pool one  
  network 192.168.10.0 255.255.255.0  
  default-router 192.168.10.1  
!  
!
```

DHCP 서버 구성

기능

- PC 자동 IP 할당
- Gateway 자동 배포

제외주소 설정

configure terminal

```
ip dhcp excluded-address 192.168.10.1  
ip dhcp excluded-address 192.168.10.100
```

DHCP Pool 설정

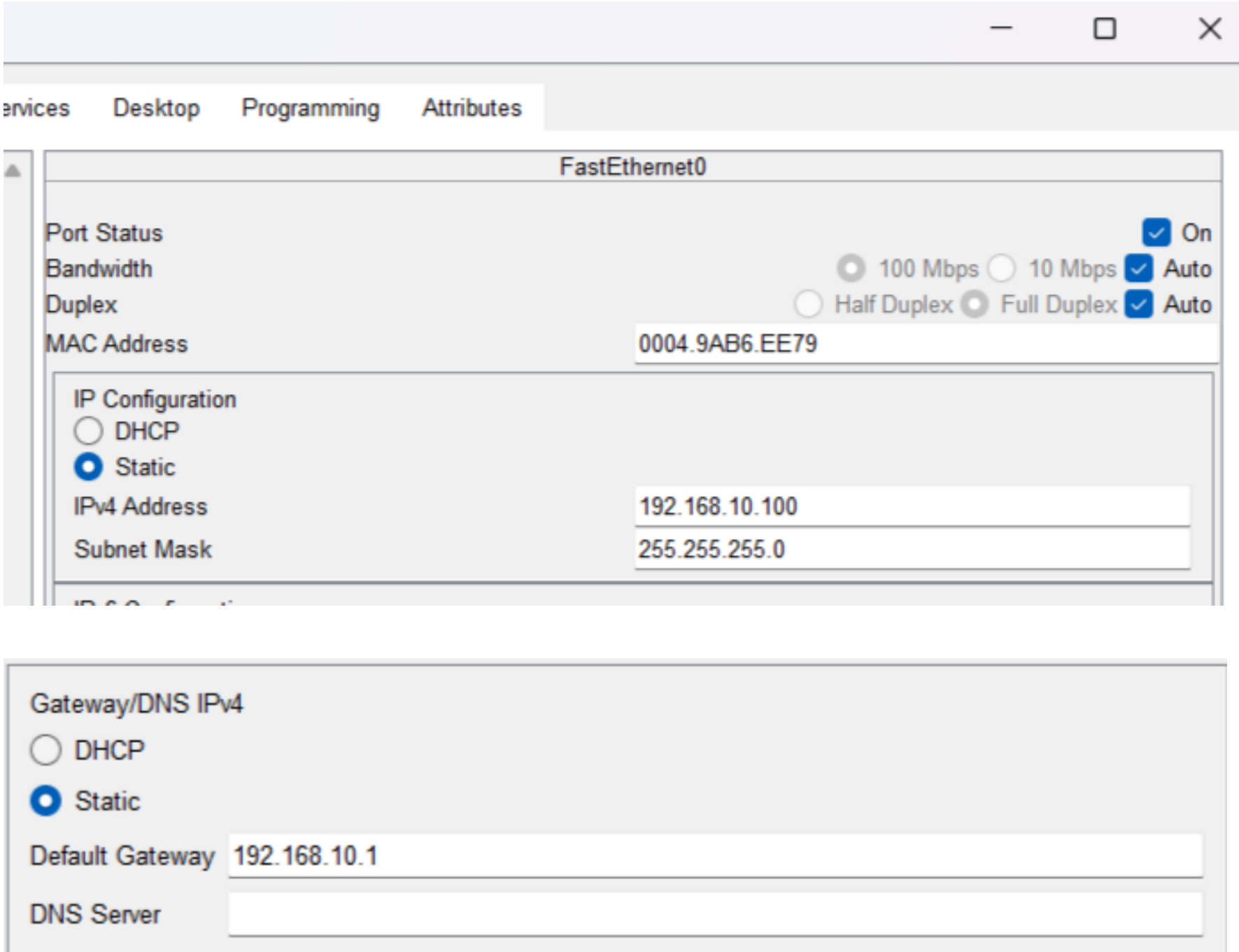
ip dhcp pool LAN

네트워크 범위 설정

```
network 192.168.10.0 255.255.255.0
```

Gateway 설정

```
default-router 192.168.10.1
```

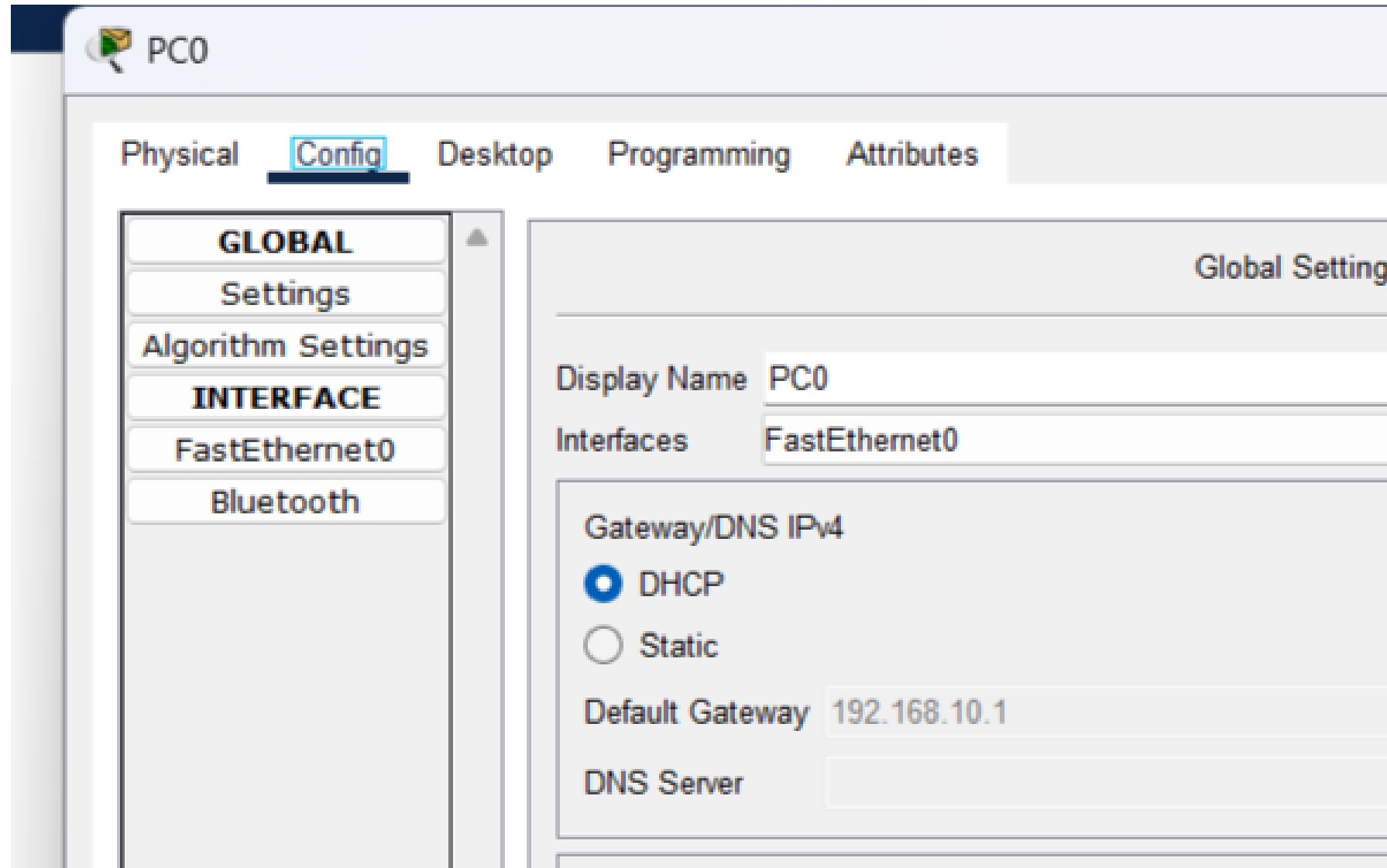



Server 고정 IP 설정

- 기능
- 서버 고정 IP 구성

설정값

항목	값
IP Address	192.168.10.100
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1



PC DHCP 설정

기능

- DHCP기반 자동 IP 설정

설정 방법

Desktop

→ IP Configuration

→ DHCP 선택

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=12ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>ping 192.168.10.100

Pinging 192.168.10.100 with 32 bytes of data:

Reply from 192.168.10.100: bytes=32 time<1ms TTL=128
Reply from 192.168.10.100: bytes=32 time<1ms TTL=128
Reply from 192.168.10.100: bytes=32 time<1ms TTL=128
Reply from 192.168.10.100: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=12ms TTL=128
Reply from 192.168.10.3: bytes=32 time=4ms TTL=128
Reply from 192.168.10.3: bytes=32 time=3ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 4ms
```

네트워크 통신 테스트

Router ping test

ping 192.168.10.1

Server ping test

ping 192.168.10.100

PC ping test

ping 192.168.10.3


```

hostname R1
!
!
!
!
!
ip dhcp excluded-address 192.168.10.1
ip dhcp excluded-address 192.168.10.100
!
ip dhcp pool one
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
!
!
!
no ip cef
no ipv6 cef
!
!
!
username admin secret 5 $1$mERr$4dpRATIgxQacPVK0CfNV4/
!
!
!
ip ssh version 2
ip domain-name lab.local
!
!
line vty 0 4
 login local
 transport input ssh
!

```

SSH 원격접속 구성

기능

- Router SSH 활성화
- Port 22 기반 원격 접속
- 사용자 인증 구성

설정 방법

configure terminal

hostname R1

exit

ip domain-name lab.local

username admin secret 1234

crypto key generate rsa > 1024

ip ssh version 2

line vty 0 4

login local

transport input ssh

end

```
C:\>ssh -l admin 192.168.10.1

Password:

R1>show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R1>
```

SSH 접속 테스트

[pc command]

ssh -l admin 192.168.10.1

> admin

>1234

검증결과

테스트 항목	결과
DHCP 자동 IP 할당	성공
PC ↔ Router Ping	성공
PC ↔ Server Ping	성공
SSH 원격 접속	성공
Router Port 활성화	성공

감사합니다.

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